A COMPREHENSIVE EVALUATION OF WIDE VIEW AND LINE FILTERS FOR EDGE DETECTION

Parameter Optimization, Benchmarking, and Noise Robustness

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OUTLINE

INTRODUCTION

The edge detection problem and motivation

THE EDGE DETECTION LANDSCAPE

Edge detection has evolved through three generations of methods.

1. Classical operators (Sobel, Prewitt, Canny): fixed-size kernels, simple but limited adaptability
2. Training-free extended filters (WVF, LF): local polynomial fitting over large neighborhoods, no learned parameters
3. Deep learning (DexiNed, TEED, PiDiNet, NBED, DiffusionEdge): learned edge representations, highest clean-data scores

Bagan and Wang's Wide View Filter (WVF) and Line Filter (LF) sit between classical simplicity and DL complexity — but are their published parameters optimal?

RESEARCH QUESTIONS

This study addresses four questions through a comprehensive four-part investigation.

1. Are the published WVF/LF parameters ($N_p = 250$, $N_s = 18$, $d = 4$) optimal for standard benchmarks?
2. How do optimally-tuned WVF/LF compare against 54 classical operators and 5 DL models?
3. Does the WVF's extended spatial support provide noise robustness advantages over learned models?
4. Are these findings statistically validated across datasets?

Total scope: 500,000+ individual filter evaluations across 330 images and 4 datasets.

METHODS

WVF and LF formulations

WIDE VIEW FILTER (WVF)

For a target pixel $\mathbf{p}$, the WVF selects N_p nearest neighbors within a circular support. At each orientation θ_k, it fits a polynomial of degree d via least squares.

The edge response is the maximum absolute normal derivative over all orientations:

$$R_{WVF}(\mathbf{p}) = \max_{k=1,\dots,N_s} |\hat{c}_2^{(k)}|$$

Three tunable parameters:

- N_p: support size (number of neighbor pixels)
- N_s: number of candidate orientations
- d: polynomial degree

LINE FILTER (LF)

The LF extends WVF by chaining $2\mathsf{m}+1$ WVF evaluations along a line at each orientation. Virtual expansion points are placed at:

$$\mathbf{v}_{\mathsf{j}}^{(\mathsf{k})} = \mathbf{p} + \mathsf{j}(\cos\theta_{\mathsf{k}}, \sin\theta_{\mathsf{k}}), \quad \mathsf{j} = -\mathsf{m}, \ldots, \mathsf{m}$$

Derivative estimates are combined via Gaussian-weighted averaging:

$$\mathsf{R}_{\mathsf{LF}}^{(\mathsf{k})}(\mathbf{p}) = \left| \sum_{\mathsf{j}=-\mathsf{m}}^{\mathsf{m}} \mathsf{w}_{\mathsf{j}} \hat{\mathsf{c}}_{2,\mathsf{j}}^{(\mathsf{k})} \right|$$

Additional parameter m (line half-width) controls tangential averaging. At $\mathsf{m} = 14$ (published), LF is $\sim$29$\times$ slower than WVF.

EXPERIMENTAL SETUP

Table: Datasets used in the evaluation (330 images total).

Dataset	Images	Resolution
UDED	30	variable
BIPED v1	50	1280×720
BIPED v2	50	1280×720
BSDS500	200	481×321

- WVF grid: 1,206 valid configurations ($N_p \times N_s \times d$)
- LF grid: 168 configurations (adding m)
- Classical baselines: 54 configurations across 7 operator families
- DL baselines: DexiNed, TEED, PiDiNet, NBED, DiffusionEdge

PARAMETER ABLATION

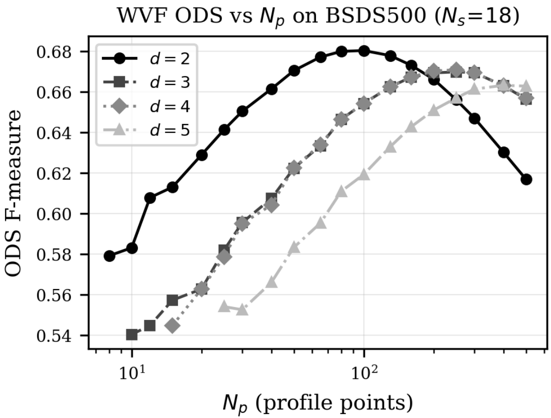
Published parameters are substantially suboptimal

WVF: SUPPORT SIZE DOMINATES PERFORMANCE

Key findings from single-image ablation:

1. ODS peaks at $N_p = 30$–50, not 250
2. N_s saturates by 4–6 orientations
3. $d = 2$ consistently outperforms $d = 4$

At published $N_p = 250$: ODS $\approx$ 0.71 versus optimal 0.86 — a 0.15 gap.

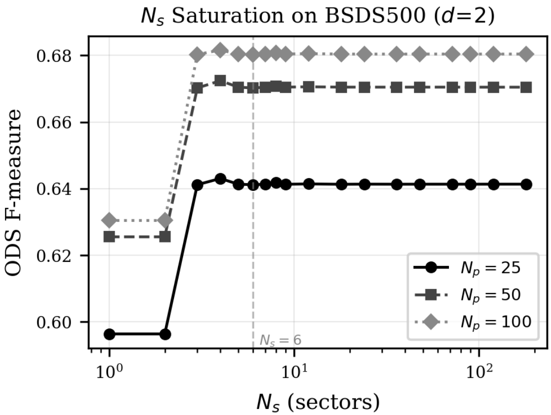


ORIENTATION COUNT SATURATES EARLY

N_s has minimal effect beyond 4–6 orientations.

The difference between $N_s = 6$ and $N_s = 120$ is typically less than 0.005 ODS.

Reducing N_s from 18 to 4 yields a $4.5\times$ speedup with no loss in edge quality.

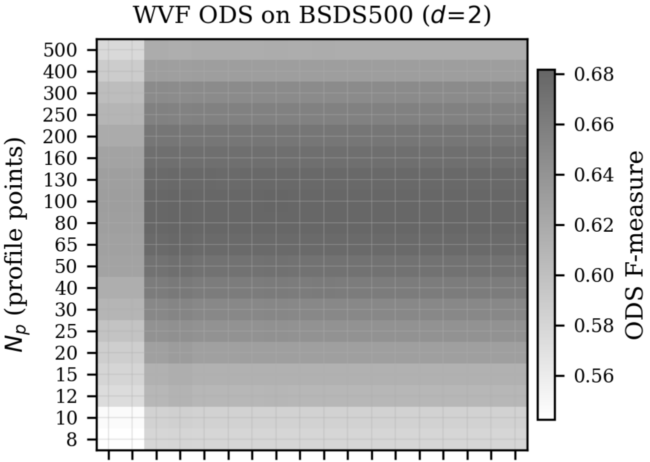


N_p VS N_s HEATMAP

The dominant variation is along N_p.

The optimal region forms a horizontal band at $N_p \approx 25–50$, largely invariant to N_s.

This confirms: support size is the critical parameter.



FULL-DATASET RESULTS: OPTIMIZED VS. PUBLISHED

Table: ODS improvement from parameter optimization across all datasets.

Dataset	Best WVF	Bagan WVF	Δ	Best LF	Bagan LF	Δ
UDED	0.899	0.847	+0.052	0.887	0.729	+0.158
BIPED v1	0.812	0.767	+0.045	0.802	0.643	+0.159
BIPED v2	0.830	0.783	+0.047	0.819	0.658	+0.161
BSDS500	0.682	0.671	+0.011	0.671	0.615	+0.056

Best WVF: $(N_p = 25, N_s = 4, d = 2)$ on three datasets. Computational savings: $112\times$ reduction over published settings.

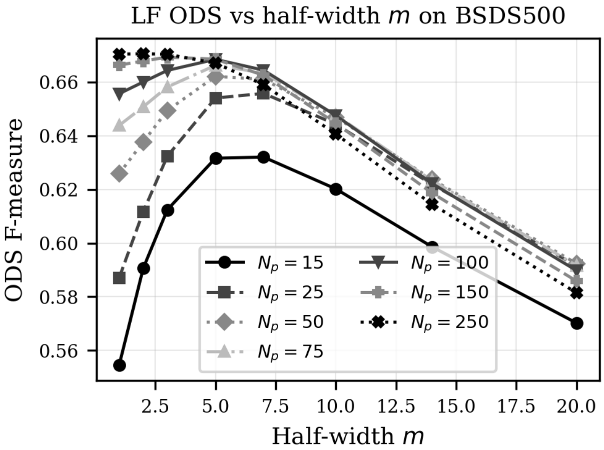
LF LINE HALF-WIDTH m: SHORTER IS BETTER ON CLEAN DATA

On all datasets, the best ODS occurs at $m = 1$–3.

The published $m = 14$ yields the worst performance.

On clean data, line-averaging harms edge localization rather than helping it.

WVF outperforms LF by 0.010–0.012 ODS on every dataset.



BENCHMARKING

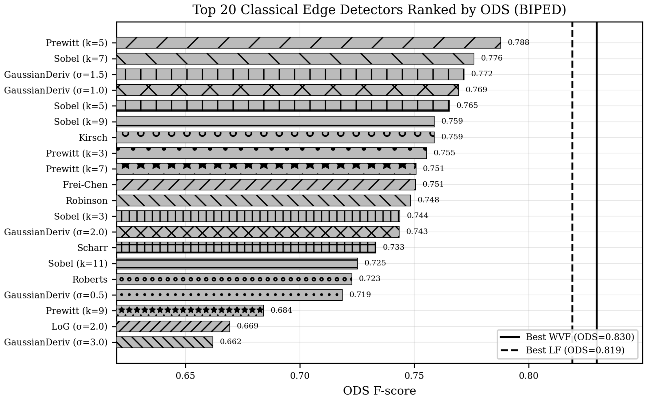
WVF vs. classical operators and deep learning

WVF OUTPERFORMS ALL 34 CLASSICAL OPERATORS

On BIPED v1:

- Best WVF: 0.812
- Best classical (Prewitt-5): 0.788
- Gap: +0.024 ODS (3.0%)

WVF exceeds every classical configuration tested, spanning Sobel, Prewitt, LoG, DoG, Kirsch, Frei-Chen, and Scharr.

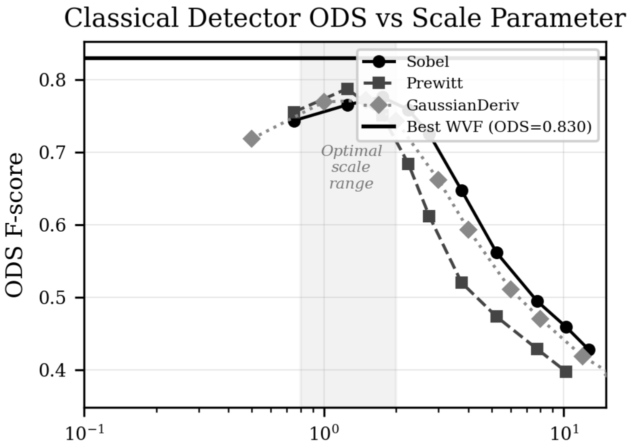


CLASSICAL SCALE SENSITIVITY

Every classical operator family shows a clear optimal scale — performance degrades sharply beyond it.

Sobel-51 scores only 0.428 versus Sobel-7 at 0.776.

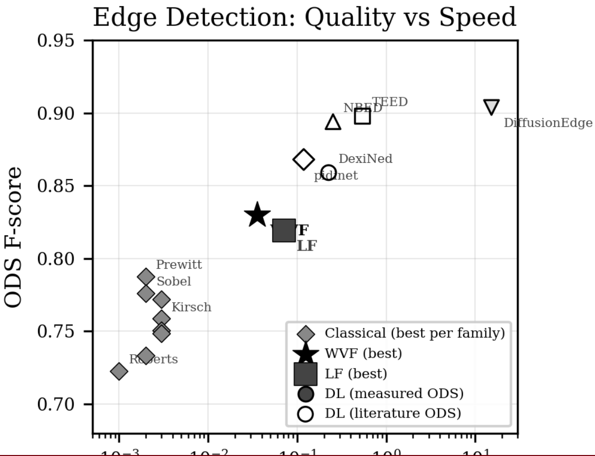
WVF's polynomial fitting performs a form of adaptive scale selection, explaining its advantage.



COMPUTATIONAL COST: WVF IS THE FASTEST METHOD

Method	Time	Rel.
WVF	0.036s	1.0×
PiDiNet	0.119s	3.3×
DexiNed	0.226s	6.3×
NBED	0.253s	7.1×
TEED	0.538s	15.1×
DiffusionEdge	15.22s	427×

WVF achieves real-time processing at 30+ fps.



NOISE ROBUSTNESS

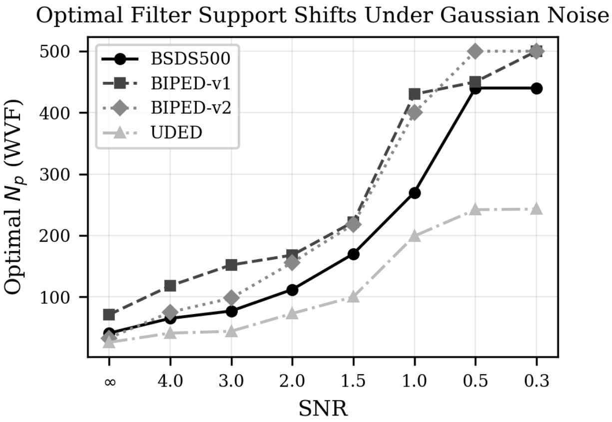
WVF adapts; deep learning collapses

OPTIMAL N_p SHIFTS DRAMATICALLY UNDER NOISE

Under severe Gaussian noise (SNR = 0.3):

- Optimal N_p shifts from $\sim$40 to $\sim$440
- A $\sim 10\times$ increase in support size
- Partially vindicates Bagan's $N_p = 250$

However, $d = 2$ remains universally optimal — $d = 4$ is suboptimal at every SNR level.

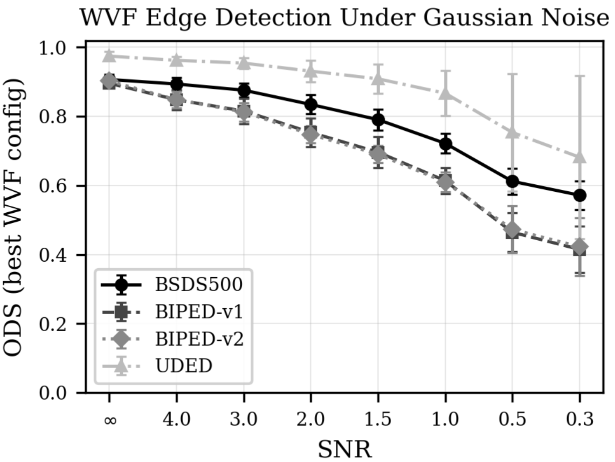


NOISE-TYPE-SPECIFIC WVF DEGRADATION

Degradation profiles vary by noise type:

- Speckle: least destructive (23% drop)
- Poisson: intermediate (30%)
- Salt-and-pepper: intermediate (31%)
- Gaussian/uniform: most severe (36–37%)

Ranking is consistent across all datasets.

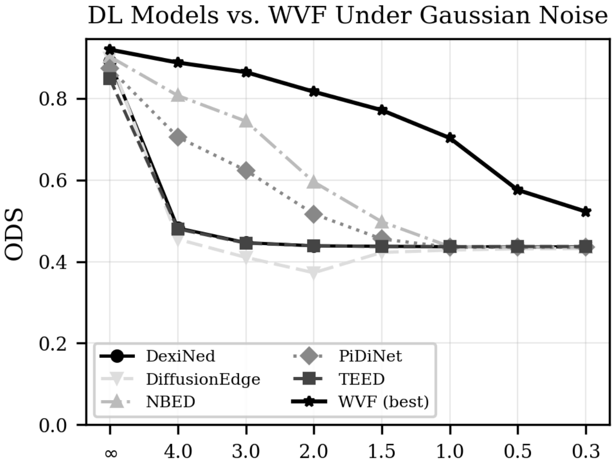


DEEP LEARNING MODELS COLLAPSE UNDER NOISE

All 5 DL models degrade severely:

- PiDiNet: 0.876 → 0.329 (62% loss)
- DexiNed: near-total collapse
- TEED: severe degradation
- NBED: most graceful decline
- DiffusionEdge: erratic behavior

None include noise augmentation in training.

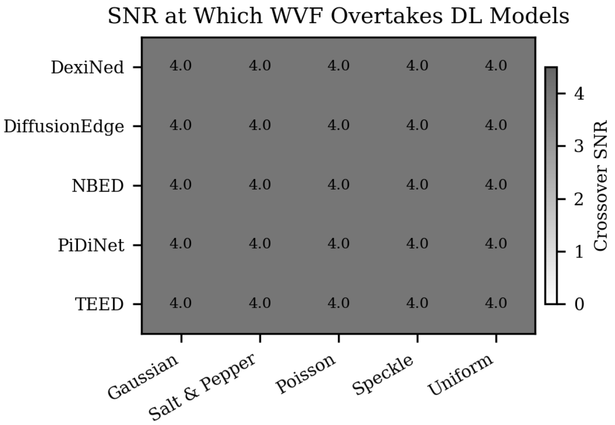


WVF OVERTAKES DL AT SURPRISINGLY MILD NOISE

Crossover analysis:

- DexiNed, TEED: overtaken at all SNR levels
- PiDiNet: overtaken at SNR $\approx$ 4.0
- NBED: most competitive, still exceeded

At SNR = 0.3: WVF beats every DL model on every test image (100% crossover).

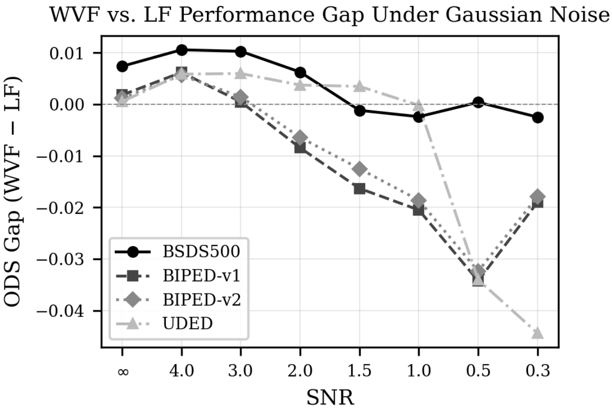


WVF VS. LF UNDER NOISE

On clean data, WVF leads by $\sim$0.007 ODS.

Under severe noise (SNR $\leq$ 1.0), the gap closes and LF slightly overtakes WVF for Gaussian, salt-and-pepper, and uniform noise.

The reversal is modest (0.002–0.009 ODS) — the LF's line-averaging provides marginal noise resilience.

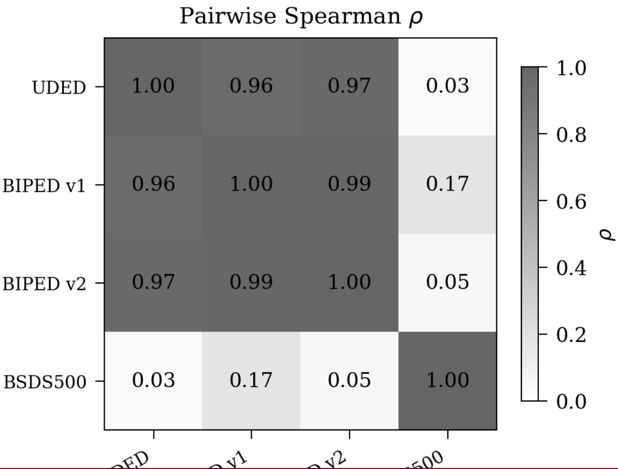


STATISTICAL VALIDATION

Nonparametric tests confirm all major claims

CROSS-DATASET RANKING CONSISTENCY

- Kendall's W $= 0.647$ (moderate concordance).
- UDED, BIPED v1, BIPED v2: $\rho > 0.96$ (near-perfect agreement).
- BSDS500 is a strong outlier: $\rho = 0.03$–0.17 with all other datasets.
- Top-10 configurations on BSDS500 share zero overlap with other datasets.

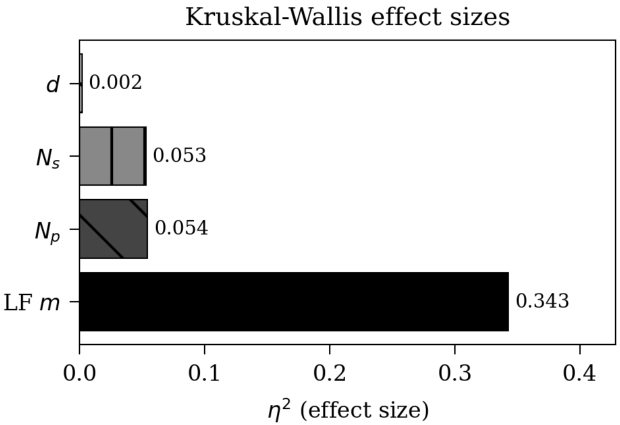


PARAMETER SENSITIVITY ANALYSIS

Kruskal–Wallis analysis reveals parameter priority:

Parameter	η^2
m (LF half-width)	0.343
N_p (support size)	$\sim$0.05
N_s (orientations)	$\sim$0.05
d (poly. degree)	$<$0.01

m is by far the most influential — the LF's line extension is its biggest liability on clean data.



CONCLUSION

SIX DEFINITIVE FINDINGS

1. Published parameters are suboptimal: $N_p = 25$–100, d $= 2$, $N_s = 4$ improves ODS by 0.01–0.16 with $112\times$ computational savings
2. WVF beats all 54 classical operators by +2.4 ODS points
3. WVF is $3.3\times$–$427\times$ faster than every DL model tested
4. Under noise, optimal N_p shifts to 250–500, partially vindicating Bagan and Wang's design
5. WVF overtakes DL models at mild noise (SNR $\approx$ 4.0); 100% crossover by SNR = 0.3
6. BSDS500 is a statistical outlier: its rankings are uninformative about other domains ($\rho = 0.03$–0.17)

PRACTICAL RECOMMENDATIONS

Clean imagery

- Use WVF (not LF)
- $N_p = 25$–50 (high-res) or 50–100 (low-res)
- $d = 2$, $N_s = 4$
- Skip NMS (raw gradient maps score higher)

Noisy environments

- Increase N_p proportionally to noise level
- Moderate noise: $N_p = 65$–112
- Severe noise: $N_p = 250$–500
- $d = 2$ remains optimal at all SNR levels

Edge detection method selection is fundamentally condition-dependent. WVF's principal strength is parametric adaptability across the full spectrum of operating conditions.

THANK YOU!

Questions & Discussion

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